FENG, RUOCHEN

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EDUCATION

Columbia University **New York City, US**
Master of Science in Data Science, GPA:3.83 / 4.00 *Aug 2025 – Expected Dec 2026*
- Courseworks: Gen AI using LLMs, Agentic AI, Causal Inference, Machine Learning, Applied Deep Learning,

University of Toronto **Toronto, CA**
Honours Bachelor of Science (H.B.Sc) with High Distinction, GPA:3.94 / 4.00 *Sep 2021 – Jun 2025*
- Program: Mathematics and Statistics Double Major, Computer Science Minor
- Scholarship & Honors: C.L. Burton Scholarship, University College Alumni Scholarship, Dean's List Scholar

WORK EXPERIENCE

Google Inc. *June 2026 – Sep 2026*
Data Scientist Intern *Greater Seattle Area, US*
- Incoming intern building and deploying **a driver-based forecasting model** for Google Maps Trips, owning the end-to-end ML pipeline from data assembly and feature engineering to modeling and evaluation.
- Developing **AI-powered forecasting tools**, including interactive dashboards, data visualization with Python and Tableau, **Gemini LLM summaries**, model interpretability (SHAP/feature importance), and scalable pipeline optimization for product-ready insights.

VeriSilicon Microelectronics Co., Ltd. *Jul 2023 – Sep 2023*
Data Engineer Intern *Shanghai, China*
- Automated a scalable financial reporting pipeline by using **AWS** (S3, Lambda), **Spark** for batch processing, and **Kafka** for data ingestion, alongside **ReportLab** and **SendGrid**, cutting generation time by ~50% and enabling cross-departmental distribution.
- Collaborated with product and engineering teams to support **A/B testing infrastructure** and **Go-To-Market analytics**, contributing to initiatives improved Small Medium-sized Business adoption by 17.3%.

OCBC Bank Ltd. *Apr 2023 – Jun 2023*
Risk Analyst Intern *Shanghai, China*
- Consolidated and cleaned fragmented financial datasets (~10M+ transaction and position records) using **SQL** techniques, including **window functions**, **CTEs**, and joins across multiple schemas, to build unified exposure tables for downstream risk analysis.
- Designed an interactive **Tableau dashboard** for real-time portfolio risk monitoring, enabling faster decision-making and higher returns.
- Built and deployed **Python-based volatility forecasting models** (e.g., **GARCH**) to assess market dynamics and quantify interest rate impacts, improving prediction accuracy by **21.4%**, with version control managed via **Git**.

RESEARCH EXPERIENCE

Deep Learning for Histopathology Image Analysis *Sep 2025 – Present*
The Reya Lab at Columbia University Irving Medical Center
- Built **supervised multi-class classification pipelines (PyTorch, scikit-learn)** across **6 annotated tissues (5 classes, 43213 tiles)**, achieving **macro F1 ~0.76–0.86** on key classes **(ADM)** through feature engineering, class balancing, and threshold tuning.
- Applied **StarDist (CNN-based instance segmentation)** on high-resolution TMA histopathology images (**7 cores, 10^5+ cells**), generating single-cell masks and spatial features; built an **interactive HTML/JS visualization** for patch-level exploration and model interpretability.

TravelMind: Domain-Specific LLM Travel Agent & Hybrid Cloud Pipeline *Sep 2025 – Dec 2025*
Supervisor: Adj Prof. Spencer Luo; Columbia University Data Science Institute & Google Inc.
- Conceptualized and designed the end-to-end system design, translating open-ended queries into a **modular multi-stage reasoning pipeline**.
- Co-designed the hybrid cloud training and deployment architecture, specifying separation of compute (**Google Vertex AI**) and artifact storage (**GCS/Hugging Face**), and contributed to implementation validation and reproducibility testing.

TECHNICAL SKILLS

- Programming Languages: Python, Java, R, SQL, MATLAB, C, C++, LaTeX, SAS, Stata
- Tools: Spark, Kafka, AWS (S3, EC2), GCP (Vertex AI), Git, Jupyter, MySQL, PostgreSQL, Tableau, Power BI, Matplotlib
- Libraries: Numpy, Pandas, PyTorch, Scikit-learn, TensorFlow, Matplotlib, RStan, tidyverse, StarDist